JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Cameras and Images
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Spring 2026

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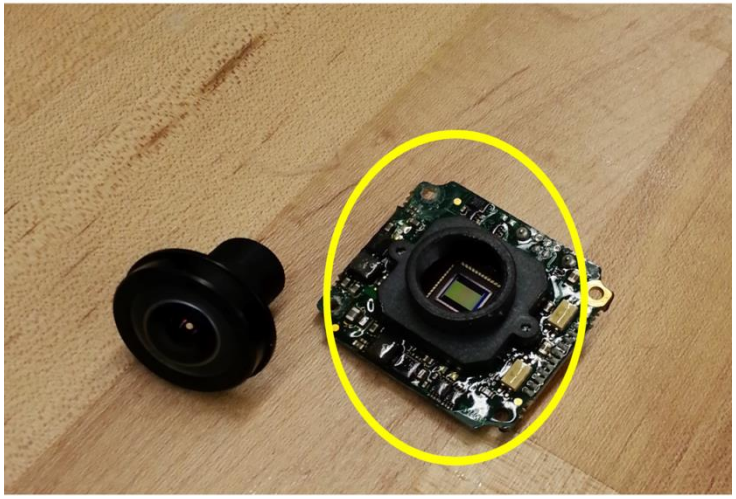
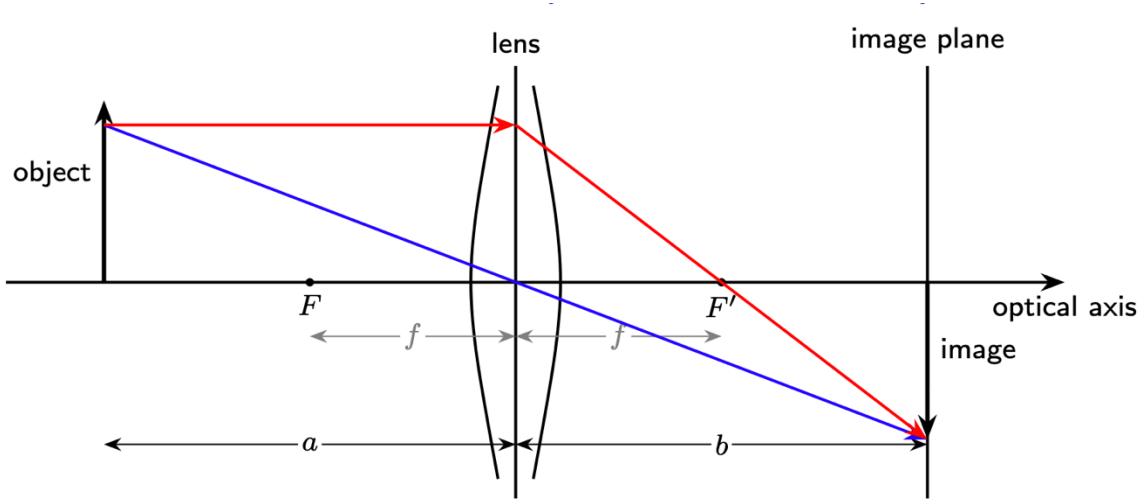
Cameras



Projection Model

Camera Projection Model



Thin lens equation

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f}$$

where a is the object distance (lens to object), b is the image distance (lens to image plane), and f is the focal length.

Focusing

$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f}$$

Auto-focus changes b and/or f to form sharp images

In-focus

out-of-focus

Perspective Camera

From the thin-lens equation

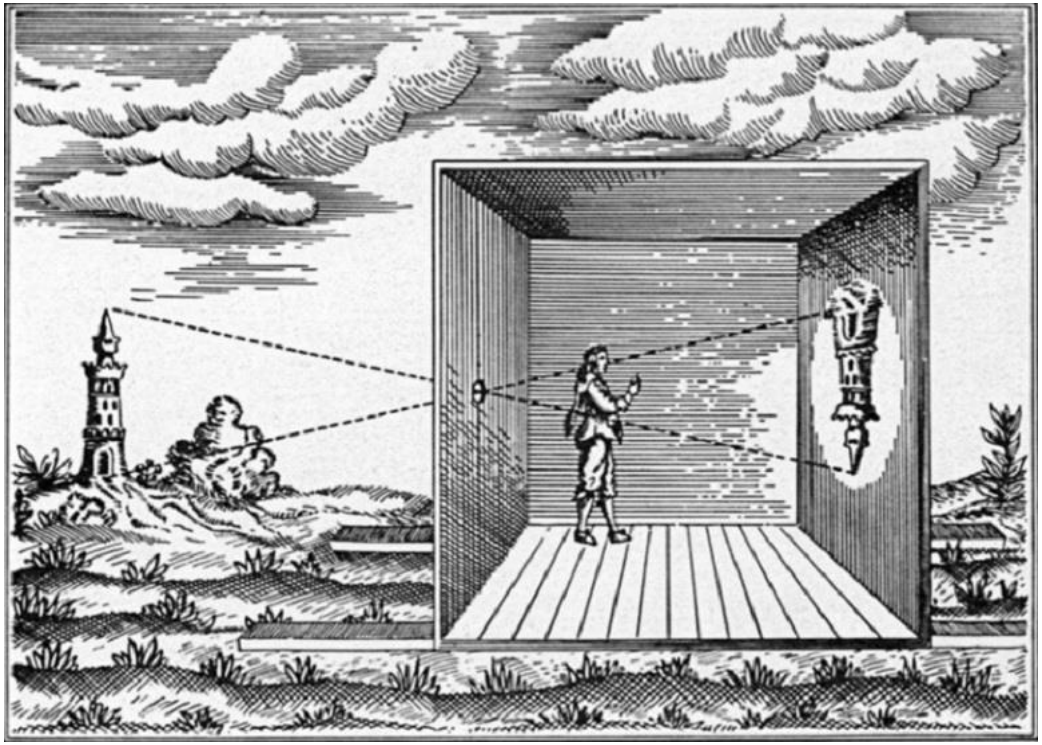
$$\frac{1}{a} + \frac{1}{b} = \frac{1}{f},$$

you get $b = f$ under the far-field (infinity-focus) assumption.

- **Assumption:** the object is very far away, $a \to \infty$ (more practically $a \gg f$).
- Then $\frac{1}{a} \approx 0$, so

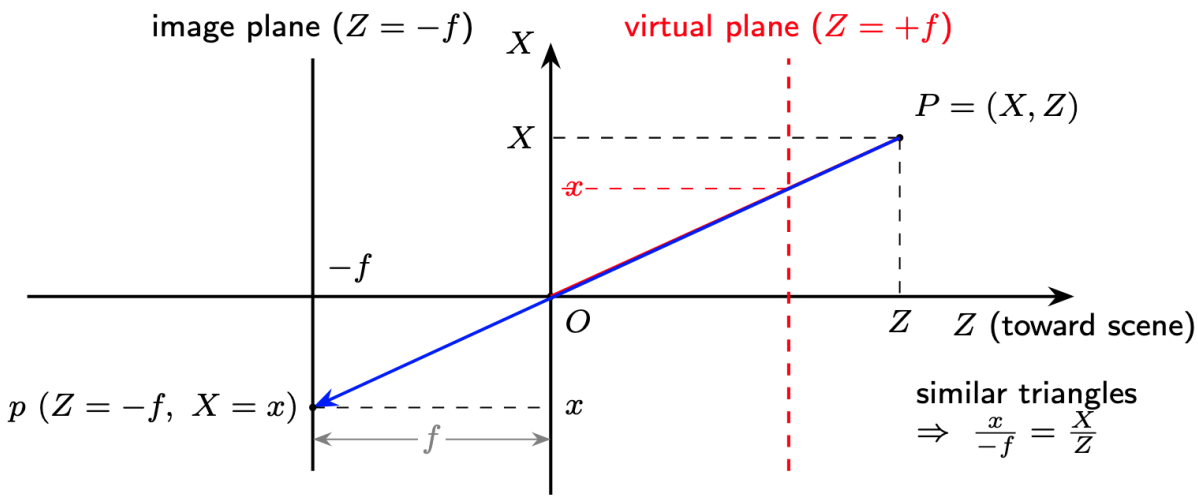
$$\frac{1}{b} = \frac{1}{f} \;\Rightarrow\; b = f.$$

First “Camera”



Camera Obscura

Pinhole Projection



Pinhole projection equation

$$x = -f\,\frac{X}{Z} \qquad \text{(physical image plane at } Z = -f\text{; image is inverted).}$$

Often we use a *virtual* image plane at $Z = +f$ to avoid the minus sign: $x = f\,\frac{X}{Z}$.

Announcement (1/28)

- Office hour changes on Canvas

Clarification about SO(3)

- Q: "Is checking the determinant = 1 sufficient to check a proper rotation matrix?"
- Answer: "**NO**. It is necessary but not sufficient."

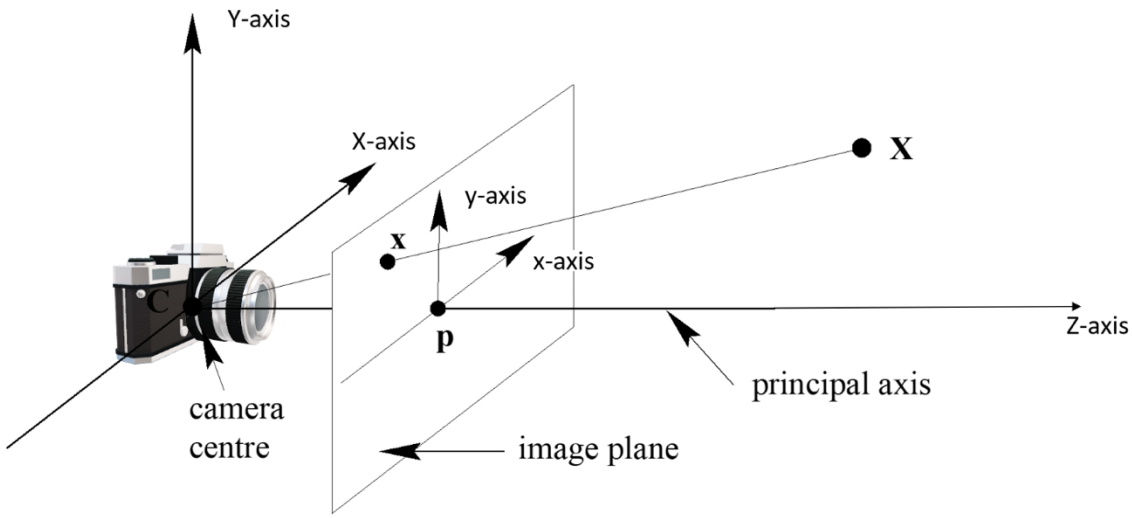
$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Shear: preserve volume, but can change the shape

For a 3×3 matrix R, a standard set of sufficient conditions for a **proper rotation** is:

$$R^\top R = I \quad \text{and} \quad \det(R) = 1.$$

Pinhole Cameras



$$x = f\frac{X}{Z}, y = f\frac{Y}{Z}$$

Metric scale (meter, inch)

For digital images, the pixel coordinates need to be mapped by (# of pixels / sensor size)

Projective Equivalence

- Projection is scale-ambiguous
- Cannot differentiable between:
  - Small, near objects
  - Big, far objects

$$\boxed{x = \frac{fX}{Z}, \qquad y = \frac{fY}{Z}.}$$

Scaling both X and Z by λ

Let $P = (X, Z)$ and define an equivalent point $P' = (X', Z')$ by

$$X' = \lambda X, \qquad Z' = \lambda Z, \qquad \lambda \neq 0.$$

Then

$$x' = f\,\frac{X'}{Z'} = f\,\frac{\lambda X}{\lambda Z} = f\,\frac{X}{Z} = x.$$

Review: Lines and Points in Euclidean Space

Lines and points in 2D

Point

A point in 2D is a vector

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2,$$

Line

A (non-vertical) line can be written implicitly as

$$ax + by + c = 0.$$

Vertical line x = 1?
a * 1 + c = 0 -> a = -c
[1, 0, -1]

It is convenient to represent the line by

$$\boldsymbol{\ell} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}, \qquad \boxed{\boldsymbol{\ell}^\top \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}} = 0.$$

Homogeneous Form of a Line

Points and Planes in 3D

Point

$$\boldsymbol{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3.$$

Plane

A plane is the set of points satisfying

$$ax + by + cz + d = 0.$$

Represent with a 4-vector

$$\boldsymbol{\pi} = \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix}, \qquad \boldsymbol{\pi}^\top \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = 0.$$

Intersecting Lines

Take two lines in Euclidean form:

$$L_1 : y = x + 1, \qquad L_2 : y = -2x + 4.$$

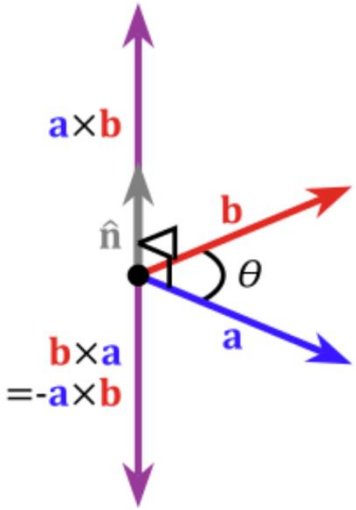
Solve in Euclidean coordinates

$$x + 1 = -2x + 4 \;\Rightarrow\; 3x = 3 \;\Rightarrow\; x = 1,\; y = 2.$$

Quick review – cross product

$$\boldsymbol{a} \times \boldsymbol{b}$$

- Output is a vector with magnitude = $||\boldsymbol{a}||||\boldsymbol{b}|| \sin \boldsymbol{\theta}_{ab}$ and direction perpendicular to both vectors.
- Right-hand thumb rule to compute direction.
- Area of the parallelogram with $\boldsymbol{a}$ and $\boldsymbol{b}$ as sides.



- Computed as the determinant $\begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$
- $\boldsymbol{a} \times \boldsymbol{b} = -(\boldsymbol{b} \times \boldsymbol{a})$

Intersecting lines via cross product

Write each line as $ax + by + c = 0$ and represent it by $\boldsymbol{\ell} = (a, b, c)^\top$:

$$L_1 : \; -x + y - 1 = 0 \;\Rightarrow\; \boldsymbol{\ell}_1 = (-1, 1, -1)^\top, \qquad L_2 : \; 2x + y - 4 = 0 \;\Rightarrow\; \boldsymbol{\ell}_2 = (2, 1, -4)^\top.$$

$$\tilde{\boldsymbol{p}} = \boldsymbol{\ell}_1 \times \boldsymbol{\ell}_2 = \begin{bmatrix} -3 \\ -6 \\ -3 \end{bmatrix} \sim \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

Can we intersect two parallel Lines?

Consider two parallel lines:

$$L_1 : y = x + 1, \qquad L_2 : y = x - 2.$$

Try to solve in Euclidean coordinates

Set the right-hand sides equal:

$$x + 1 = x - 2 \;\Rightarrow\; 1 = -2,$$

which is a contradiction. So there is **no solution** $(x, y) \in \mathbb{R}^2$.

No finite intersection

Points at infinity

- A “new” geometry allows us to compute the intersection of two parallel lines at a point at infinity

Exercise: find intersection between two lines

$$L_1 : y = x + 1, \qquad L_2 : y = x - 2.$$

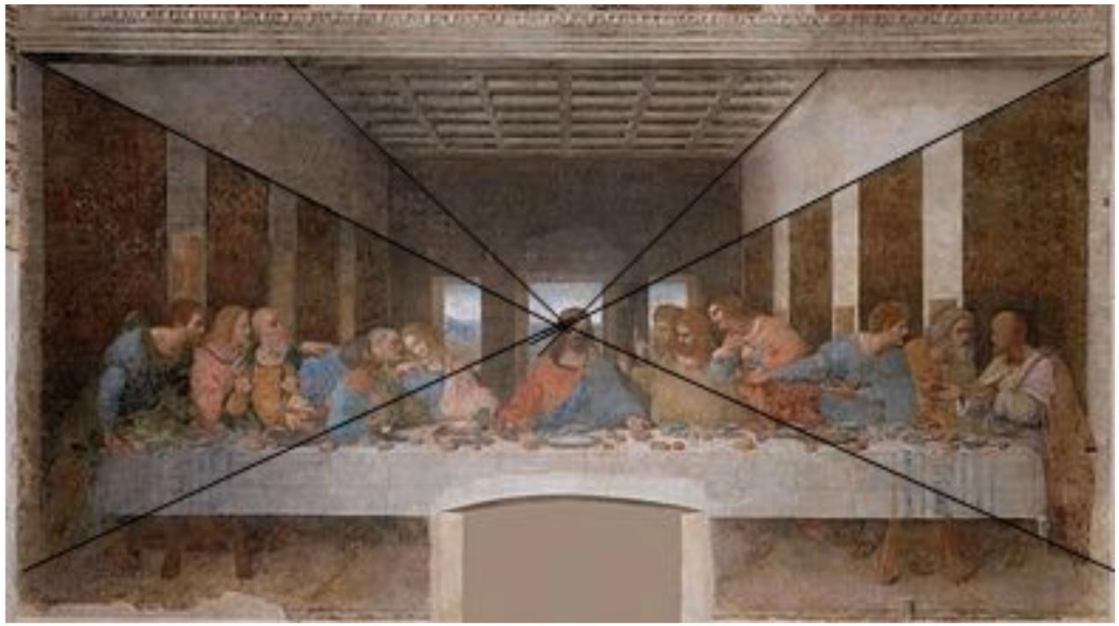
W = 0 -> points at infinity

Projective Geometry

When parallel line intersect

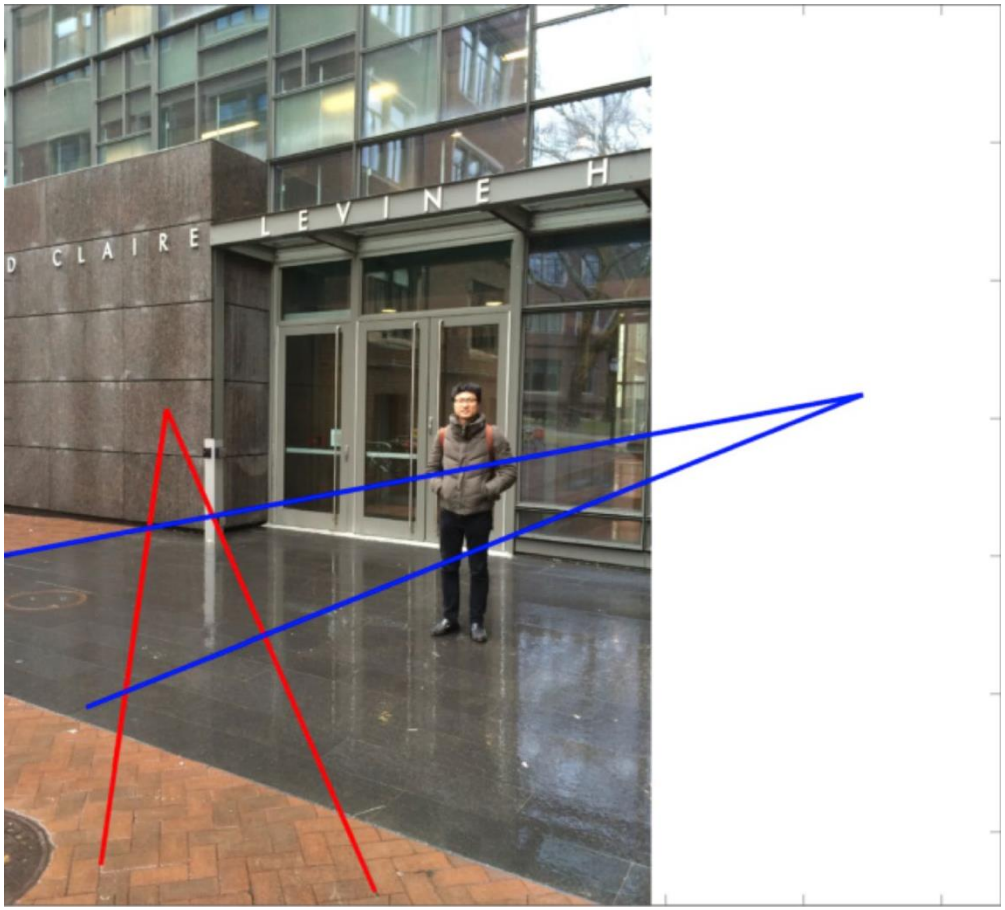
Vanishing Points

- Parallel lines intersect in perspective renderings



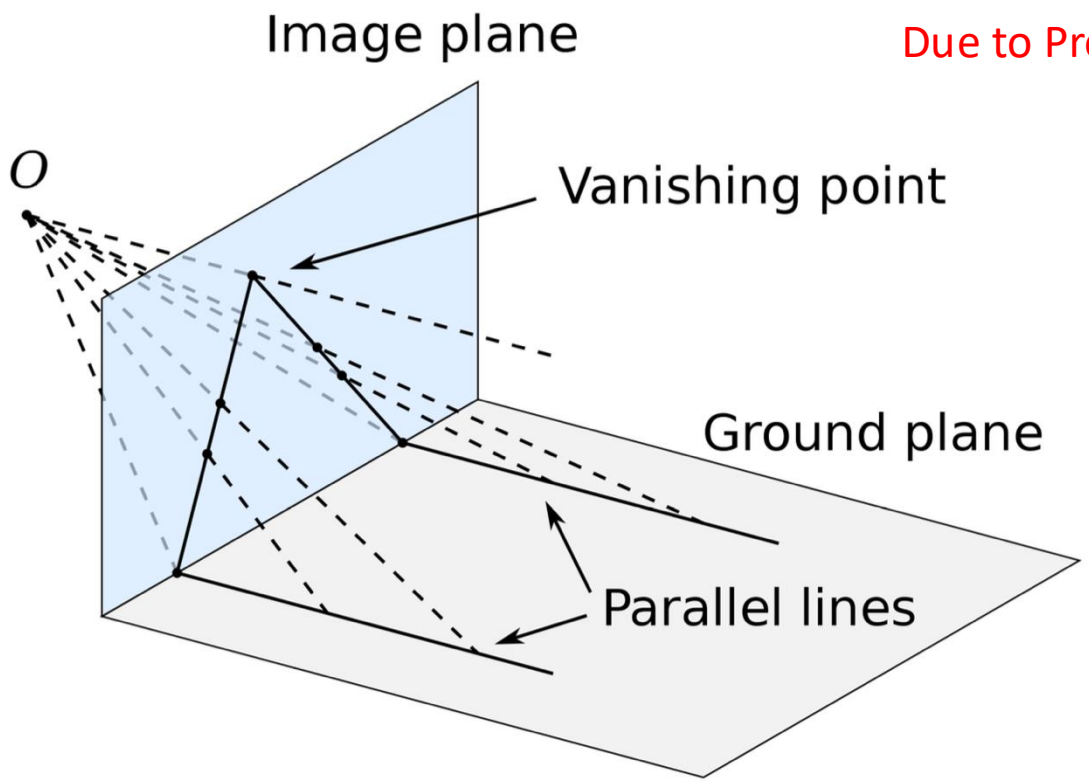
Da Vinci's "Last Supper"

Vanishing Points



AI Generated Videos

Where vanishing points are from



Due to Projection

Projective vs Euclidean Geometry

Homogeneous coordinates: $\mathbb{R}^n \rightarrow \mathbb{P}^n$

- Extension of Euclidean geometry to include points at infinity

- Why?
  - We want to handle points at infinity (no special cases)
  - Projective relationships are natural
    - e.g., 3D points along a pixel ray all map to the same 2D point

Homogeneous Coordinates

A Euclidean point in $\mathbb{R}^n$ maps to homogeneous coordinates in $\mathbb{P}^n$ by appending a 1:

$$(x_1, \ldots, x_n)^\top \in \mathbb{R}^n \quad \longmapsto \quad (x_1, \ldots, x_n, 1)^\top \in \mathbb{P}^n.$$

Recover Euclidean coordinates

If $w \neq 0$ for $\boldsymbol{u} = (u_1, \ldots, u_n, w)^\top$, then

$$(x_1, \ldots, x_n) = \left(\frac{u_1}{w}, \ldots, \frac{u_n}{w}\right).$$

$$\begin{bmatrix} x \\ y \end{bmatrix} \xrightarrow{\text{homogeneous coords}} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

We will normally use w to denote this homogeneous vector element

Notation

Projective equivalence

For nonzero vectors $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^{n+1} \setminus \{\mathbf{0}\}$, we write

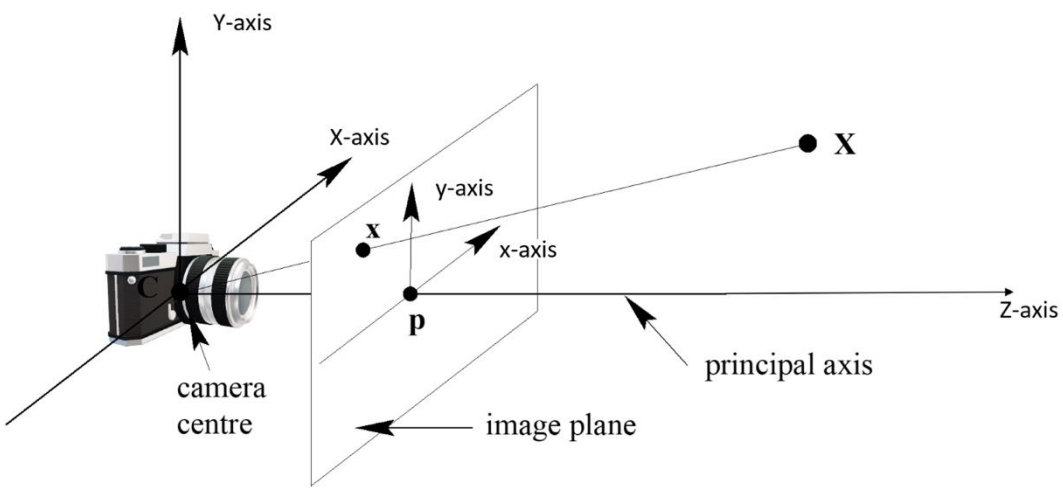
$$\boldsymbol{u} \sim \boldsymbol{v} \quad \Longleftrightarrow \quad \exists\, \lambda \neq 0 \text{ s.t. } \boldsymbol{v} = \lambda \boldsymbol{u}.$$

Examples in $\mathbb{P}^2$

$$\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} \sim \begin{bmatrix} -3 \\ -6 \\ -3 \end{bmatrix}, \qquad [1:2:1] = [2:4:2] = [-3:-6:-3].$$

A projective point is just a ray

- A point is a ray from the projection center to the physical 3D point
- Normalization (w=1) means the point location when depth is 1 (along the ray)



Example: 2D point in homogenous coordinates

Take a Euclidean point in $\mathbb{R}^2$:

$$\bar{\boldsymbol{x}} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Homogeneous embedding in $\mathbb{P}^2$

$$\boldsymbol{x} \sim \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}.$$

Scale-equivalent representations (same projective point)

Any nonzero scale represents the same point:

$$\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 4 \\ -2 \\ 2 \end{bmatrix} \sim \begin{bmatrix} -6 \\ 3 \\ -3 \end{bmatrix}.$$

Connecting two points

Pick two points in homogeneous coordinates:

$$\boldsymbol{p}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad \boldsymbol{p}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Compute the line through them

$$\boldsymbol{\ell} = \boldsymbol{p}_2 \times \boldsymbol{p}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \sim \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

This line is $\boldsymbol{\ell}^\top \boldsymbol{x} = 0$, i.e.

$$x + y - 1 = 0 \quad \text{(in affine coordinates } \boldsymbol{x} = (x, y, 1)^\top\text{)}.$$

Check: $\boldsymbol{\ell}^\top \boldsymbol{p}_1 = 0$ and $\boldsymbol{\ell}^\top \boldsymbol{p}_2 = 0$.

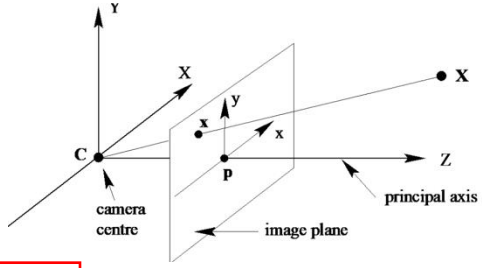
Cross product in projective geometry

- Line and line -> point
- Point and point -> line

Perspective projection becomes linear

In Euclidean coordinates, perspective projection is nonlinear due to division by Z:

$$x = \frac{fX}{Z}, \qquad y = \frac{fY}{Z}.$$



In homogeneous coordinates, it is linear:

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}, \qquad \boxed{(x, y) = \left(\frac{u}{w}, \frac{v}{w}\right).}$$

Make sure the last coordinate w is 1

*Later we will see a more general version of this

What about w = 0?

- Homogeneous coordinates are defined up to nonzero scale:

$$(x, y, 1) \sim (2x, 2y, 2) \sim (wx, wy, w).$$

- If $w = 0$, then $(u, v, 0)$ corresponds to a *point at infinity*.
- Geometric picture: rays parallel to the image plane represent points at infinity.

Intuition: divide by zero is infinity

Geometric meaning: the common direction of all lines in (u, v) direction

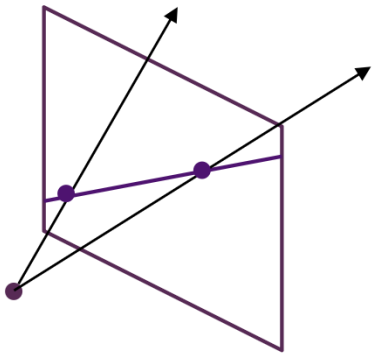
Projective Lines

Line equation in $\mathbb{P}^2$

A line is represented by $\boldsymbol{\ell} = (a, b, c)^\top$ and satisfies

$$\boldsymbol{\ell}^\top \boldsymbol{x} = 0, \qquad \boldsymbol{x} = (x, y, w)^\top \in \mathbb{P}^2.$$

Geometrically, a line in the image corresponds to a plane of rays through the origin.



Exercise

- Where do image lines x=1 and x=2 intersect?

Two vertical lines: $x = 1$ and $x = 2$

$$x = 1 \;\Rightarrow\; \frac{X}{W} = 1 \;\Rightarrow\; X - W = 0 \;\Rightarrow\; \boldsymbol{\ell}_1 = (1, 0, -1)^\top.$$

$$x = 2 \;\Rightarrow\; \frac{X}{W} = 2 \;\Rightarrow\; X - 2W = 0 \;\Rightarrow\; \boldsymbol{\ell}_2 = (1, 0, -2)^\top.$$

Intersection via cross product

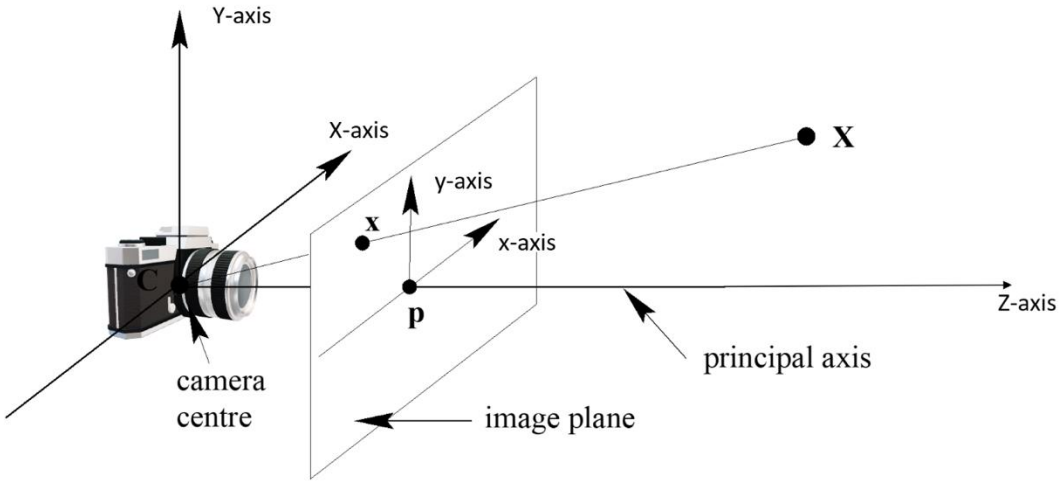
Their intersection is

$$\tilde{\boldsymbol{p}} = \boldsymbol{\ell}_1 \times \boldsymbol{\ell}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \sim [0 : 1 : 0].$$

How to project points

Geometric Intuition of projection

- Draw a ray from the center to the point
- A projection is to find out "where does the ray hit the image plane?"



Step 1: compute ray (homogeneous)

Given $P = (X, Y, Z)$ with $Z > 0$, the viewing ray is

$$\boldsymbol{r}(\lambda) = \lambda \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}, \qquad \lambda > 0.$$

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \boldsymbol{r}\left(\frac{1}{Z}\right) = \frac{1}{Z} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \boxed{\begin{bmatrix} \frac{X}{Z} \\ \frac{Y}{Z} \\ 1 \end{bmatrix}}$$

Normalized coordinate

Step 2: apply intrinsics matrix

General intrinsics (principal point, pixel scaling, optional skew)

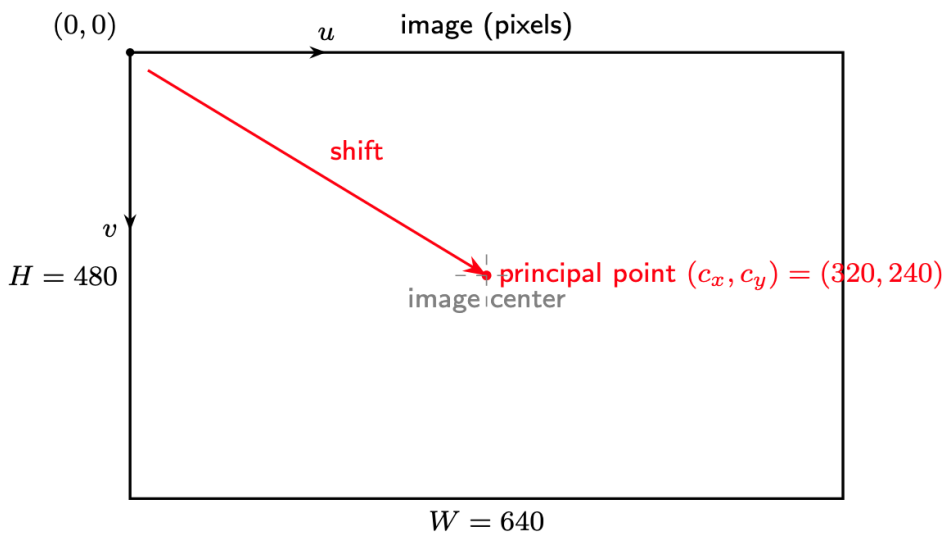
$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{(focal lengths + principal point).}$$

Move the point to the actual focal length
Center of image is not (0, 0)

We get this matrix through camera calibration

Intrinsic matrix: principal point

Pixel coordinates typically use an origin at the **top-left** corner, while the optical axis intersects the image plane at the **principal point** (c_x, c_y).



So even if the normalized point is (x, y), the pixel is shifted: $u = f_x x + sy + c_x$, $v = f_y y + c_y$.

Step 2: apply intrinsics matrix

General intrinsics (principal point, pixel scaling, optional skew)

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \qquad \text{(focal lengths + principal point).}$$

Principal point and focal lengths in pixels

With principal point (c_x, c_y) and focal lengths (f_x, f_y) (in pixels), the mapping is

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \boldsymbol{r}\left(\frac{1}{Z}\right) = \frac{1}{Z}\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} \frac{X}{Z} \\ \frac{Y}{Z} \\ 1 \end{bmatrix}$$

$$u = f_x x + c_x, \qquad v = f_y y + c_y$$

Combined in one step

$$\tilde{\boldsymbol{p}} \sim K\,\boldsymbol{X}_c \iff \begin{bmatrix} u \\ v \\ w \end{bmatrix} \sim \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = \begin{bmatrix} f_x X + sY + c_x Z \\ f_y Y + c_y Z \\ Z \end{bmatrix}$$

$$(u_{\mathsf{px}}, v_{\mathsf{px}}) = \left(\frac{u}{w}, \frac{v}{w}\right)$$

Full Camera Matrix

General intrinsics (principal point, pixel scaling, optional skew)

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

- f_x, f_y: focal length measured in *pixels* (includes pixel size / aspect ratio)
- (c_x, c_y): **principal point** (projection of optical axis) in pixel coordinates
- s: skew (often ≈ 0)

Exercise: Point projection

Point in camera frame

$$\boldsymbol{P}_c = (X, Y, Z)^\top = (1,\ 0.5,\ 4)^\top, \qquad f = 800.$$

Homogeneous projection (linear in $\mathbb{P}$)

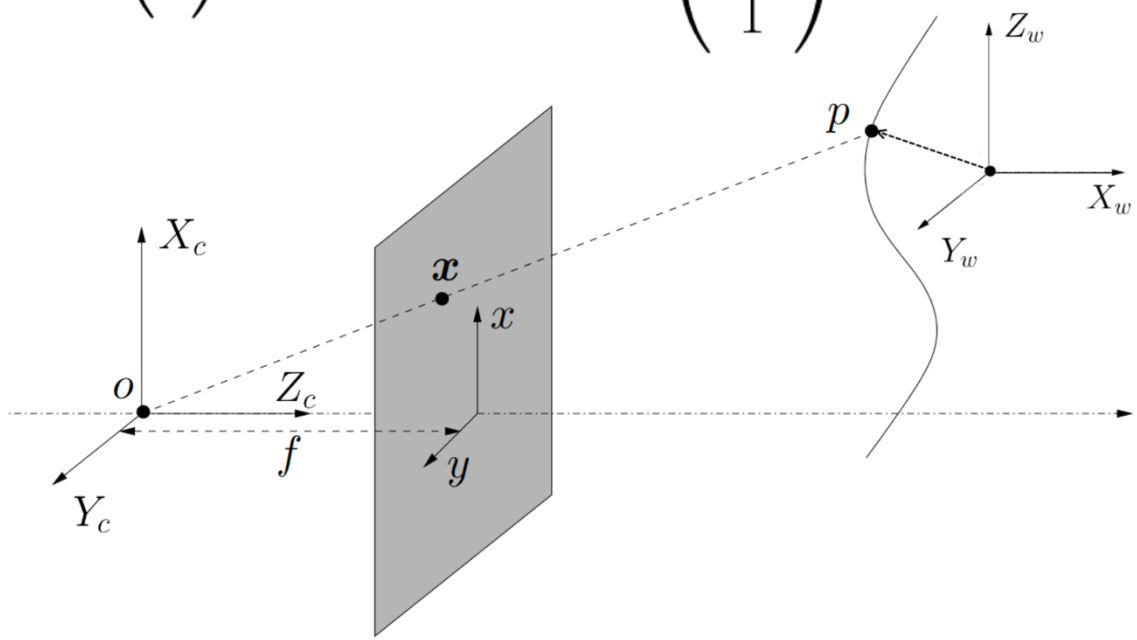
$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \underbrace{\begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}}_{P} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \begin{bmatrix} 800 \\ 400 \\ 4 \end{bmatrix}.$$

Dehomogenize

$$(x, y) = \left(\frac{u}{w}, \frac{v}{w}\right) = \left(\frac{800}{4}, \frac{400}{4}\right) = (200, 100).$$

Projection of world points

$$\lambda \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = K \begin{pmatrix} {}^cR_w & {}^cT_w \end{pmatrix} \begin{pmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{pmatrix}$$



Projection of world points

Now assume the point is given in a **world frame** as $\boldsymbol{X}_w = (X_w, Y_w, Z_w)^\top$.

Extrinsics (camera pose)

The camera-frame coordinates are

$$\boldsymbol{X}_c = R_w^c \, \boldsymbol{X}_w + \boldsymbol{t}_w^c, \qquad R_w^c \in SO(3), \; \boldsymbol{t}_w^c \in \mathbb{R}^3.$$

In homogeneous form:

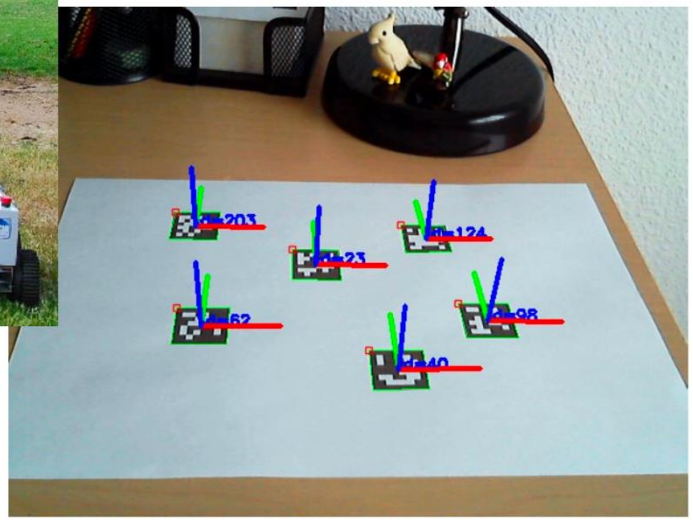
$$\begin{bmatrix} \boldsymbol{X}_c \\ 1 \end{bmatrix} = \begin{bmatrix} R_w^c & \boldsymbol{t}_w^c \\ \boldsymbol{0}^\top & 1 \end{bmatrix} \begin{bmatrix} \boldsymbol{X}_w \\ 1 \end{bmatrix}.$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \boxed{\underbrace{\begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}}_{K} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \end{bmatrix}}_{[R\,|\,\boldsymbol{t}]}} \begin{bmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{bmatrix}.$$

3 x 4 Projection Matrix

Applications of Projective Geometry

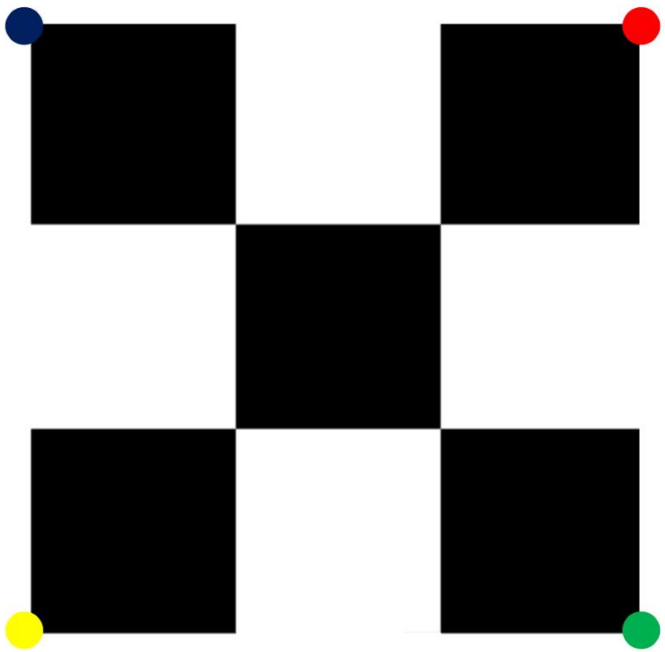
Application: April Tags



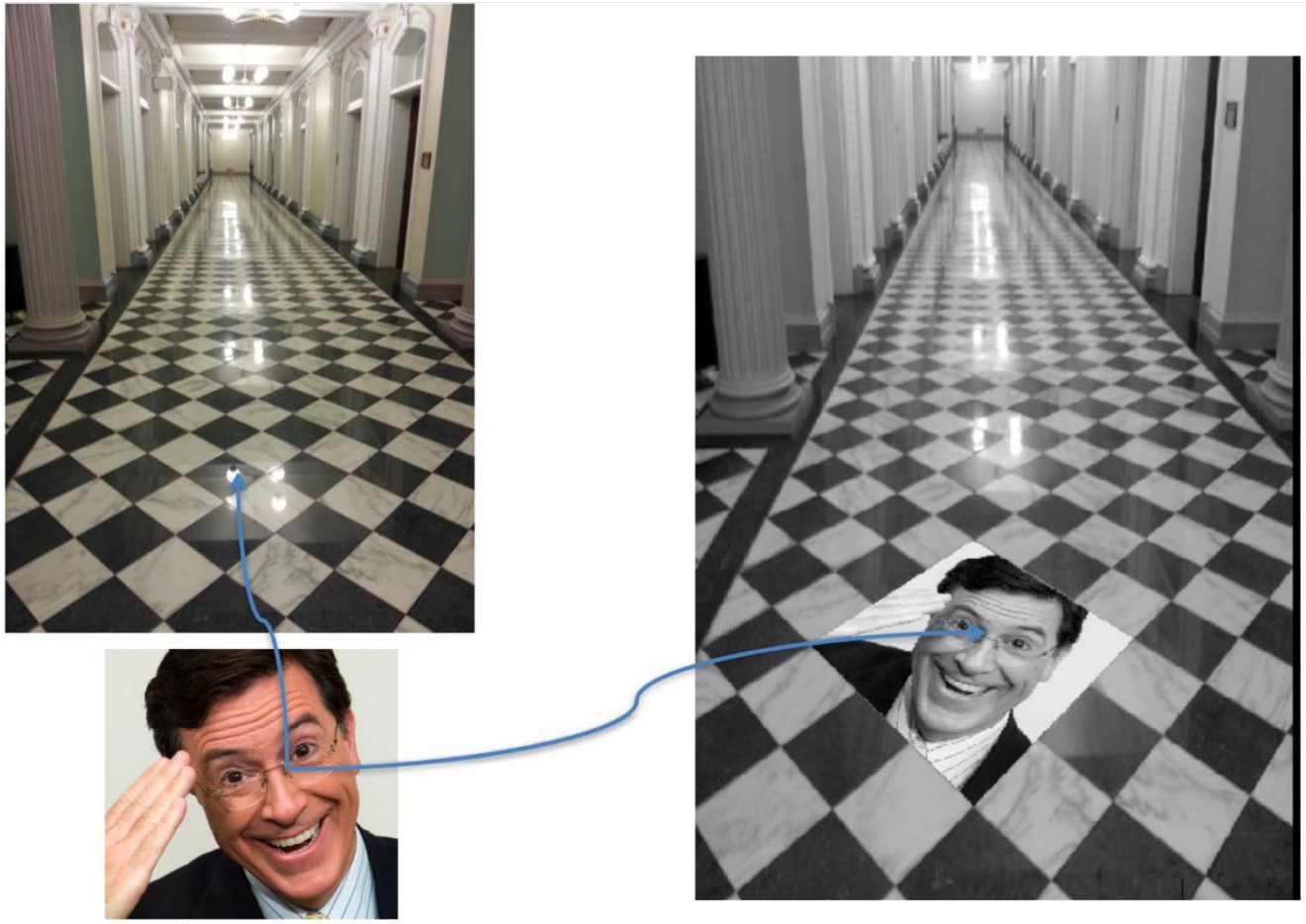
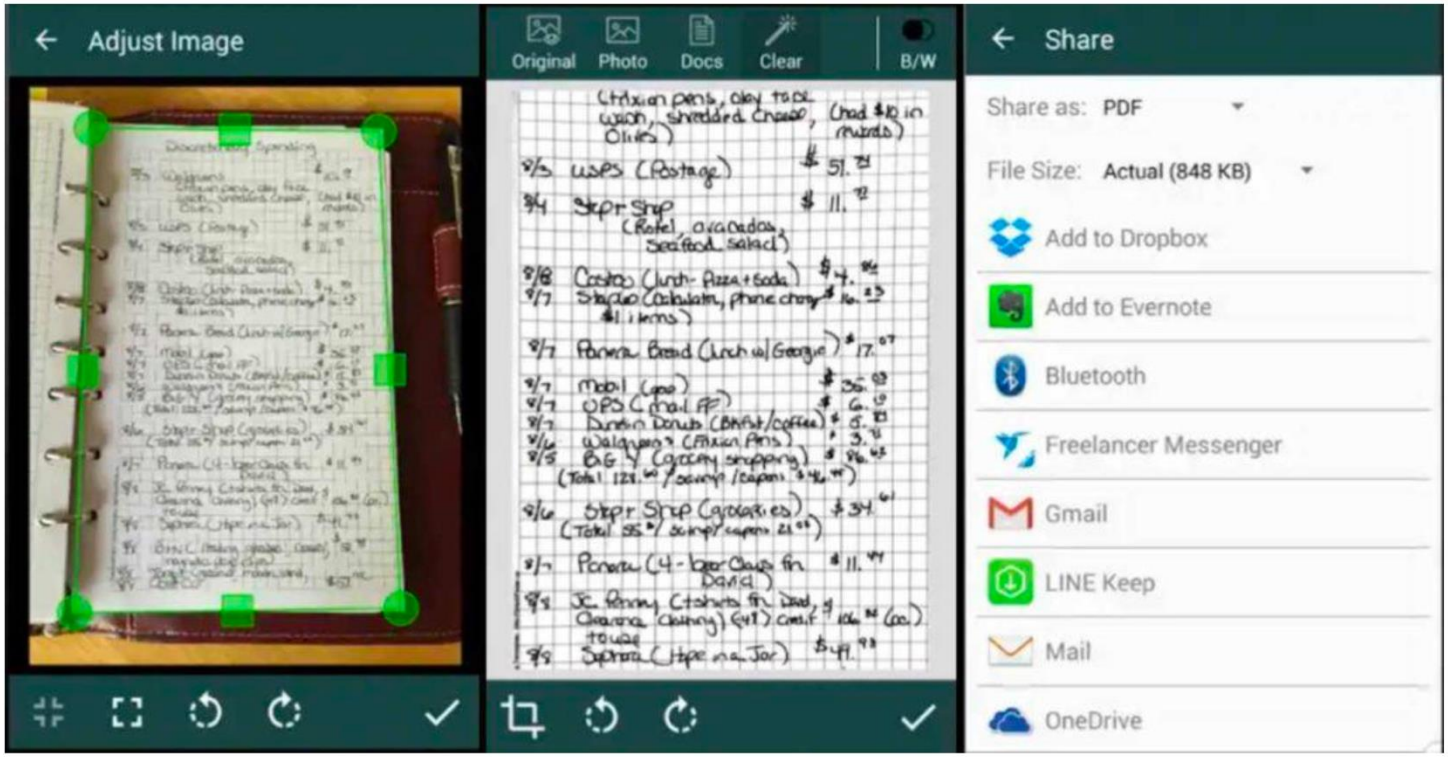
Applications: Panoramas

Compute a transform

If we want to compute the transformation between a reference image and a new image



Applying a transform



Document Scanner